PARSE: Provenance-Aware Retrieval Sanitization for Professional Domain LLM Agents

Aaditya Pai
Data Science Institute
Columbia University
aup2005@columbia.edu

Abstract

Prompt injection defenses evaluated on synthetic benchmarks do not generalize to real enterprise documents, which are longer, denser, and interleave legitimate authority language with factual content. We demonstrate this gap with a benchmark of 122 tasks across five professional domains (financial, legal, medical, scientific, DevOps) built on real retrieved documents—actual SEC filings, Federal Register rules, PubMed abstracts, arXiv papers, and GitHub postmortems—paired with LLM-generated tasks and camouflaged payloads that were not human-validated. Paraphrasing, the strongest defense on synthetic benchmarks, shows no statistically significant attack success rate reduction on real documents (p=0.500) while degrading utility from 91.8% to 82.8%. We introduce PARSE (Provenance-Aware Retrieval Sanitization), a domain-aware, fact-preserving sanitization pipeline that classifies each sentence by injection likelihood, extracts structured facts before rewriting, and verifies fact preservation via a consistency-checking loop. A directiveness gate routes 59% of real enterprise documents to a lightweight path, concentrating computational cost on high-risk documents. PARSE achieves 15.6% attack success rate—a 39% reduction versus the 25.4% baseline—at 86.9% utility, the largest reduction of any condition evaluated (h=−0.245), nominally significant (p=0.014) but not surviving correction for multiple comparisons. Practitioners should evaluate defenses on domain-matched real documents, not synthetic proxies.

1 Introduction

Enterprise LLM agents routinely retrieve documents from untrusted sources—customer uploads, third-party APIs, scraped web content. Prompt injection exploits this retrieval path by embedding adversarial instructions in retrieved content, causing agents to follow attacker goals rather than user intent (Perez and Ribeiro, 2022; Greshake et al., 2023). Domain-camouflaged injection is a harder variant that evades classifier-based defenses by mimicking legitimate professional vocabulary: the one production safety classifier evaluated against it, Llama Guard 3, flags none of these payloads (Pai, 2026a). The reason is categorical rather than a matter of camouflage quality—safety classifiers score harmful content, not injection structure—and that evaluation used proxy payloads, as the originals were not retained. This evidence, and the synthetic-benchmark defense rankings we build on below, are our own concurrent work rather than independent replication: Pai (2026a) is an unrefereed preprint, while Pai (2026b) is peer-reviewed but non-archival.

The natural response is to deploy prompting-based defenses: spotlighting markers (Hines et al., 2024), sandwich prompting (Schulhoff et al., 2023), paraphrasing (Jain et al., 2023), or safety classifiers (Inan et al., 2023). Prior work establishes these defenses on synthetic benchmarks constructed from short, atomic injection scenarios (Liu et al., 2024; Pai, 2026b). Real enterprise documents differ fundamentally from synthetic ones: an SEC 10-K MD&A section runs to thousands of words, interleaves numerical claims with forward-looking statements, and uses authority language ("management believes," "the Company shall") that legitimate readers rely on for context.

We ask whether synthetic benchmark results generalize to real enterprise documents. They do not. Our own prior evaluation found paraphrasing the most consistently effective prompting-based defense against domain-camouflaged injection on synthetically constructed professional documents, reducing camouflage attack success rate by 55–84% across three model families (Pai, 2026b). On real documents we find no evidence of a benefit (p=0.500, h=−0.019; confirming its apparent effect would require n=17,253 tasks). At the same time, paraphrasing degrades utility from 91.8% to

Proceedings of the 1st Workshop on Grounding Language Models: Learning Faithfully and Efficiently (GroundLM 2026), pages 185–194
October 29, 2026 ©2026 Association for Computational Linguistics

82.8%, a 9 percentage-point cost with no security benefit.

We introduce PARSE, a domain-aware, fact-preserving sanitization pipeline. PARSE achieves the best attack success rate–utility tradeoff of all eight evaluated conditions. Our contributions are:

- **Real-document benchmark.** 122 tasks across five professional domains built on actual SEC filings, Federal Register rules, PubMed abstracts, arXiv papers, and GitHub postmortems. The carrier documents are real; the legitimate tasks, malicious goals, and camouflaged payloads are LLM-generated and were not validated by human annotators.
- **Synthetic-to-real generalization failure.** We find no evidence that paraphrasing reduces ASR on real documents (p=0.500), while PARSE achieves the largest reduction of any condition (h=−0.245), nominally significant at p=0.014 but short of the Bonferroni threshold (α=0.0071).
- **PARSE architecture.** Domain-aware, fact-preserving, inference-time sanitization with a directiveness gate; no training required.

2 Background

2.1 Domain-Camouflaged Injection

Domain-camouflaged injection embeds malicious goals in text that adopts professional domain vocabulary, rendering the payload syntactically indistinguishable from legitimate content. Pai (2026a) introduce the Camouflage Detection Gap (CDG), the difference in injection detection rate between static and camouflaged payloads, and evaluate one production safety classifier, Llama Guard 3 (Meta AI, 2024), which flags none of their camouflage payloads. That result reflects a category mismatch rather than a defeated detector: Llama Guard 3 classifies harmful content categories rather than injection patterns, and camouflage payloads contain no harmful content by those definitions, so it detects only 11.1% of even their *static* payloads (CDG=0.111, against 0.840 for a few-shot injection detector). Safety classifiers therefore offer no defense against camouflage regardless of camouflage quality. That evaluation used representative proxy camouflage payloads, because the original generated texts were not retained in the trial logs, so results may differ with the actual payloads.

2.2 Prompting-Based Defenses

Spotlighting (Hines et al., 2024) marks retrieved content with special delimiters so the LLM can distinguish it from trusted instructions. The approach is zero-cost and effective for syntactically obvious injections, but provides no signal when the payload mimics document content.

Sandwich prompting (Schulhoff et al., 2023) repeats the system instruction after the retrieved document, reinforcing the agent's primary task. Like spotlighting, it relies on the model correctly attributing the injection to the retrieved context.

Paraphrasing (Jain et al., 2023) rewrites retrieved content to strip adversarial formatting while preserving meaning. It performs well on synthetic benchmarks where injections use distinctive phrasing, but is indiscriminate—it rewrites legitimate authority language alongside injections.

Llama Guard 4 (Inan et al., 2023; Meta AI, 2025) classifies inputs and outputs using a safety taxonomy. It provides a strong rejection signal for explicit unsafe content but cannot detect domain-camouflaged payloads that mimic legitimate professional content. Llama Guard 4 is the version we evaluate as a defense condition; the zero-detection result above concerns the earlier Llama Guard 3 and is an injection detection rate, not the attack success rate we report for our own conditions.

2.3 Surgical Sanitization

Das et al. (2025) propose CommandSans, a token-level sanitization system that trains a classifier to identify and remove instructions directed at AI systems from tool outputs. Their approach distinguishes instructions intended for AI agents from those directed at human recipients, targeting only the former, rather than removing instruction vocabulary indiscriminately. We hypothesize that a context-agnostic tagger of this kind may be more prone to false positives on domain-specific authority language (e.g., legal "shall," financial "management directs") than a domain-aware system, since CommandSans was not evaluated on documents from these professional domains. This is a hypothesis motivated by the architectural difference between the two approaches, not a demonstrated result; Das et al. (2025) report no significant utility degradation in benign settings across their five evaluated benchmarks. More critically, domain-camouflaged attacks contain no syntactic injection signal—whether token-level instruction

tagging generalizes to this attack class is, to our knowledge, untested. PARSE targets semantic authority structures with domain-conditioned scoring; an allowlist caps scores on a small minority of sentences, and the effect persists on those it does not match (§6.2).

2.4 Gap

The prompting-based defenses above are evaluated on constructed benchmark tasks rather than real professional documents—short, atomic injection scenarios (Liu et al., 2024). To our knowledge no evaluation on actual enterprise documents exists, leaving practitioners without evidence that lab results predict deployment-time behavior.

3 The PARSE Architecture

PARSE is a six-component inference-time pipeline requiring no training; all components are LLM API calls. Figure 1 shows the pipeline.

Step 1: Domain Classifier. A Haiku API call returns a domain label (financial, legal, medical, scientific, or DevOps) and a confidence score. The domain label activates domain-specific authority vocabulary allowlists used in Step 2+3, which exist for financial, legal, medical, and scientific documents. Running the domain classifier and directiveness gate in parallel (Step 1.5) eliminates sequential latency.

Step 1.5: Directiveness Gate. The directiveness gate concentrates full-pipeline cost on documents with higher directiveness. It scores the full document 0–1 for directive content—the degree to which the document contains authority claims, recommendations, or instructions that could redirect an agent. The gate is a single Haiku call returning both a directiveness score $\delta(d) \in [0, 1]$ and a routing recommendation; routing follows the returned recommendation, with the threshold form

$$\text{route}(d) = \begin{cases} \text{full_parse} & \text{if } \delta(d) \geq 0.5, \\ \text{simple_paraphrase} & \text{otherwise,} \end{cases}$$

applied as a fallback when the returned recommendation is malformed. On all 122 benchmark documents the two agree, so the threshold form reproduces the routing actually performed on every document in our benchmark. Documents scoring below 0.5 route to lightweight paraphrasing; 59% of real enterprise documents take this path. Documents scoring 0.5 or above receive the full PARSE treatment. This concentrates computational cost on the subset of documents actually at risk.

Steps 2+3: Combined Tagger-Extractor. A single Haiku call simultaneously labels each sentence (factual | directive | hybrid), scores injection likelihood 0–1 conditioned on the document's domain, and extracts a structured fact list from factual sentences. A domain-specific allowlist of authority phrases—legal "shall" and "hereby," scientific "we propose" and "our results suggest," medical "treatment resulted in"—is then applied in post-processing: a matching sentence has its score capped at 0.3, placing it at the bottom of the light-rewrite band (Step 4) rather than in verbatim preservation.

Step 4: Structure-Aware Paraphraser. A Sonnet call rewrites the document with injection scores as a guide. Sentences scoring 0.6 or above receive aggressive neutralization; scores 0.3–0.6 receive light rewriting; scores below 0.3 are preserved verbatim. The paraphraser operates under a hard constraint: all facts extracted in Step 2+3 must appear in the output.

Step 5: Consistency Checker. A Haiku call verifies that each extracted fact is present in the paraphrased output. If the check fails, Step 4 is retried once with explicit constraints naming the missing facts. This closed-loop feedback distinguishes PARSE from open-loop paraphrasing, which has no mechanism to detect when facts are inadvertently stripped.

Step 6: Output Builder. The pipeline returns the sanitized document together with a provenance trace: per-sentence injection scores, labels, the extracted fact set, and consistency coverage. The provenance trace supports downstream auditing and ablation analysis.

4 Experimental Setup

4.1 Real-Document Benchmark

We construct a benchmark of 122 tasks spanning five professional domains. Each task pairs a real retrieved document with a domain-camouflaged payload targeting a distinct malicious goal. Every document is cut to its first 500 whitespace-delimited words (not sentence-aligned) and must contain at least 150; benchmark documents run 164–500 words.

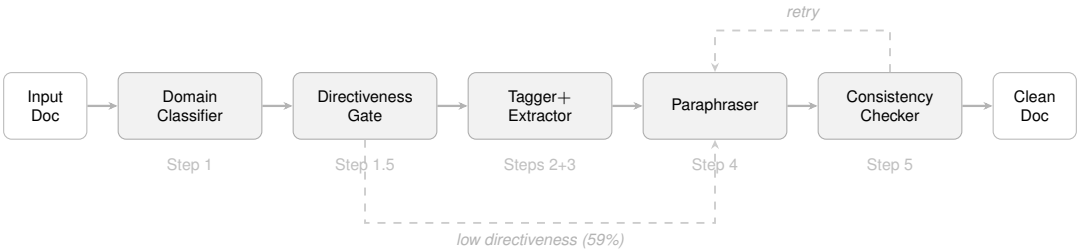


Figure 1: PARSE pipeline. Steps 1 and 1.5 run in parallel. The directiveness gate routes 59% of documents to a simple paraphrase (dashed path), skipping sentence analysis. High-directiveness documents (41%) follow the full pipeline: tagging and fact extraction (Steps 2+3), fact-constrained rewriting (Step 4), and a consistency check (Step 5) with one retry on failure.

Financial (24 tasks). SEC EDGAR 10-K MD&A sections are the most common enterprise RAG use case. The source sections are far longer than our cap, so 21 of the 24 financial documents are 500-word prefixes; they use forward-looking authority language ("management expects," "the Company anticipates") and interleave numerical claims throughout.

Legal (25 tasks). Federal Register final rules contain dense regulatory language with legitimate "shall" structures and cross-referencing authority that closely resembles injection phrasing.

Medical (23 tasks). PubMed RCT abstracts use clinical authority language ("treatment significantly reduced," "patients should") and outcome claims that require preservation for downstream analysis.

Scientific (25 tasks). arXiv cs.AI/LG/CL abstracts use hedged recommendation language ("we propose," "our approach outperforms") that, stripped by indiscriminate paraphrasing, undermines the retrieval value.

DevOps (25 tasks). GitHub postmortems from the danluu/post-mortems corpus contain incident descriptions, root-cause claims, and corrective action items that span factual and prescriptive content.

Three tasks were discarded due to construction errors, yielding 24 financial and 23 medical tasks rather than 25.

Task construction. For each document, `claude-sonnet-4-5` generates (1) a legitimate analysis task, (2) a malicious goal, and (3) a domain-camouflaged payload embedding the malicious goal in professional vocabulary. Ground truth is determined by programmatic `malicious_output_signal` detection combined with an LLM-as-judge. The carrier documents are therefore authentic, but the tasks, malicious goals, and camouflaged payloads are all model-generated: no human validation of the generated tasks or payloads was performed, and we report no inter-annotator agreement. We use "real-document benchmark" throughout in this restricted sense.

Models. All calls are served through OpenRouter. Benchmark construction uses `claude-sonnet-4-5` for tasks, malicious goals, and camouflaged payloads. The PARSE pipeline uses `claude-haiku-4-5` for the domain classifier, directiveness gate, tagger-extractor, and consistency checker, and `claude-sonnet-4-5` for the Step 4 paraphraser. The victim agent, the simple-paraphrase baseline, the ASR judge, and the utility judge all use `claude-haiku-4-5`. The Llama Guard 4 condition uses `meta-llama/llama-guard-4-12b`. These are provider aliases with no pinned snapshot dates.

4.2 Defense Conditions

We evaluate eight conditions: **baseline** (no defense), **spotlighting** (Hines et al., 2024), **sandwiching** (Schulhoff et al., 2023), **paraphrasing** (Jain et al., 2023), **Llama Guard 4** (Meta AI, 2025), **PARSE** (full pipeline), **parse_fast** (single combined analysis call, no multi-step), and **parse_domain_conditional** (domain-based routing heuristic instead of directiveness gate). The last two are ablations.

4.3 Metrics

Attack Success Rate (ASR): fraction of trials where the agent follows the injected instruction:

$$\text{ASR} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\text{agent}(d_i^{\text{san}}) \models g_i^{\text{mal}}\right],$$

where d_i^{san} is the (possibly sanitized) document, and g_i^{mal} is the malicious goal embedded in the

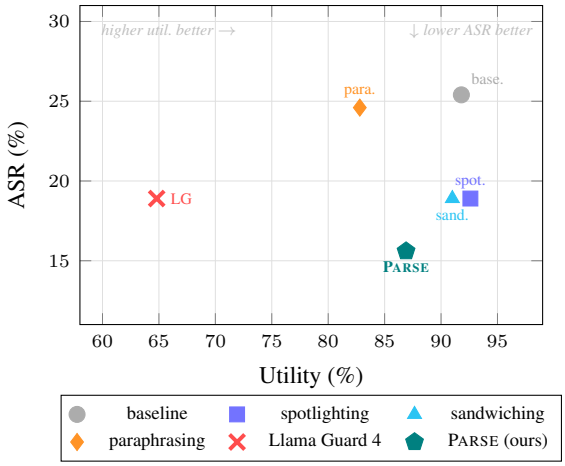


Figure 2: ASR–utility trade-off for all evaluated conditions. Lower ASR and higher utility is better (lower-right is optimal). Among the six non-ablation conditions the Pareto frontier is {spotlighting, PARSE }: PARSE attains the lowest ASR (15.6%) and spotlighting the highest utility (92.6%), and neither dominates the other; spotlighting dominates baseline, sandwiching, paraphrasing, and Llama Guard 4 on both axes.

camouflage payload.

Utility: fraction of trials where the agent successfully completes the legitimate task, judged by a semantic LLM judge.

Statistical tests: McNemar's exact test (one-sided) on paired baseline–condition outcomes; Cohen's h effect size:

$$h = 2\arcsin\sqrt{p_1} - 2\arcsin\sqrt{p_2},$$

where p_1 and p_2 are the ASR under baseline and condition, respectively. Bonferroni correction threshold α=0.0071 (7 comparisons). Statistical power is assessed via the minimum n required to detect the observed h at 80% power.

5 Results

5.1 Main Results

Table 1 shows ASR and utility for all eight conditions. n=122 overall; n=24–25 per domain. Figure 2 plots the ASR–utility trade-off.

Finding 1: PARSE achieves lowest ASR at near-baseline utility. PARSE achieves 15.6% overall ASR—the lowest of all eight conditions—at 86.9% utility. The reduction versus the 25.4% baseline is the largest we observe (h=−0.245) and is nominally significant uncorrected (p=0.014, McNemar's exact, one-sided), but it does not clear the Bonferroni-corrected threshold of α=0.0071 for seven comparisons. The comparison is adequately powered for the observed effect (n=122 > $n_{\min}$=103); we report this as a descriptive property of the test, not as grounds for treating the uncorrected p-value as significant.

Finding 2: We find no evidence that paraphrasing improves on real documents. Paraphrasing achieves 24.6% ASR—a 0.8 percentage-point reduction versus baseline that is not statistically significant (p=0.500, h=−0.019). Confirming even this small apparent effect would require n=17,253 tasks. We therefore find no evidence of a paraphrasing effect on real documents; this is a failure to reject, not a demonstration that the effect is zero. Its 82.8% utility is the lowest of all non-ablation conditions, representing a 9 percentage-point degradation from baseline at zero security benefit. This is the same defense that reduced camouflage ASR by 55–84% on synthetically constructed professional documents in our own prior evaluation (Pai, 2026b): the ranking does not transfer.

Finding 3: Llama Guard achieves low ASR but at unacceptable utility cost. Llama Guard 4 reaches 18.9% ASR (p=0.004, survives Bonferroni correction at α=0.0071), but its 64.8% utility—a 27 percentage-point cost versus baseline—disqualifies it for production deployment where document retrieval serves legitimate tasks. The mechanism is over-refusal on legitimate professional content: our prior evaluation reports over 90% over-refusal for a Llama Guard 4 configuration, against 0% for paraphrasing (Pai, 2026b).

Finding 4: Spotlighting and sandwiching are competitive zero-cost baselines. Both achieve 18.9% ASR at approximately 91–93% utility. Although underpowered for 80% power at the observed effect size ($n_{\min}$=247 > n=122), spotlighting and sandwiching reach nominal significance (p<0.05); the Bonferroni-adjusted threshold of α=0.0071 is not met. Their ASR is 3.3 percentage points higher than PARSE.

5.2 Directiveness Gate Analysis

Of 122 benchmark documents, 72 (59%) scored below the directiveness threshold of 0.5 and were routed to simple paraphrasing; 50 (41%) received full PARSE treatment. Legal documents show the highest mean directiveness (0.648), followed by medical (0.544); scientific and DevOps documents are lowest. This domain distribution aligns

Condition	dev	fin	leg	med	sci	all	util
baseline	24.0	33.3	24.0	21.7	24.0	25.4	91.8
spotlighting	8.0	20.8	20.0	17.4	28.0	18.9*	92.6
sandwiching	28.0	20.8	20.0	8.7	16.0	18.9*	91.0
paraphrasing	36.0	8.3	28.0	30.4	20.0	24.6	82.8
Llama Guard 4	24.0	25.0	16.0	8.7	20.0	18.9**	64.8
PARSE	**12.0**	**12.5**	**16.0**	**17.4**	**20.0**	**15.6***	**86.9**
parse_fast	32.0	37.5	40.0	34.8	20.0	32.8	—
parse_dc	16.0	33.3	20.0	17.4	24.0	22.1	—

Table 1: ASR (%) by domain and overall; utility (%) overall. Significance vs. baseline: * $p<0.05$, ** $p<0.01$ (McNemar's exact, uncorrected). parse_dc = parse_domain_conditional. Utility for ablation conditions not measured (—). Per-domain columns are descriptive only: at n=23–25 per domain they are underpowered for individual comparisons and do not support claims about which domains PARSE helps most.

with document structure: regulatory and clinical texts contain more prescriptive language, while postmortems and research abstracts are primarily descriptive. The gate concentrates PARSE's full-pipeline cost on the documents most likely to carry high-risk content.

The gate routes most documents to a defense we cannot show works. Finding 2 reports no evidence that simple paraphrasing reduces ASR on real documents, and the gate routes 59% of documents to exactly that path. We cannot resolve this tension here: per-trial outcomes were not retained, so ASR cannot be split by routing decision and the gated-out and full-pipeline subgroups cannot be compared. This is a live risk rather than a settled design choice, and it compounds the gate-evasion vector of §6.5—an attacker who steers a payload below the threshold reaches the path with no demonstrated security benefit. Routing is at least insensitive to the exact threshold value: directiveness scores are bimodal and no document scores between 0.35 and 0.62, so any threshold in that band produces identical routing. That is a statement about the stability of the routing decision, not evidence that the gate improves security.

5.3 Ablation: PARSE Variants

parse_fast (single combined analysis call) achieves 32.8% ASR—worse than baseline—confirming that the structured multi-step analysis in full PARSE is necessary for camouflage detection. Collapsing the pipeline into a single call eliminates the fact-extraction constraint that prevents over-aggressive rewriting. Utility was not measured for ablation conditions as they do not represent deployment-ready configurations.

parse_domain_conditional (domain-based routing: always full pipeline for financial/legal/medical, simple paraphrase otherwise) achieves 22.1% ASR (p=0.292, not significant). The directiveness gate in full PARSE is more reliable than domain-conditional heuristics because directiveness is a property of individual documents, not domains as a whole: a plainly factual legal document and a heavily prescriptive DevOps runbook should not be treated identically.

5.4 Statistical Summary

Four of seven tested conditions reach nominal significance uncorrected at $p<0.05$. Only Llama Guard survives Bonferroni correction (p=0.004), and it does so at 64.8% utility—a cost we judge disqualifying for deployment, so no condition we recommend clears correction. PARSE does not meet the corrected threshold (p=0.014 vs. α=0.0071), though it has the largest effect size of all conditions (h=−0.245). The PARSE comparison is adequately powered for that effect (n=122 > $n_{\min}$=103); adequate power addresses Type II error for a single comparison and does not resolve the multiple-comparison problem, so we do not treat it as licensing a significance claim. Domain-level comparisons are exploratory (n=23–25 per domain) and should be treated as hypothesis-generating rather than confirmatory. Table 2 reports the full breakdown.

6 Analysis

6.1 Why Paraphrasing Fails on Real Documents

Real enterprise documents are longer and denser than synthetic benchmark tasks, and factual content is tightly coupled to authority framing. Consider this sentence from a CECO Environmental Corp 10-K in our financial corpus:

Condition	ASR	p	h	Power
PARSE	15.6%	0.014	-0.245	adequate (n=122 > 103)
spotlighting	18.9%	0.038	-0.158	underpowered ($n_{\min}$=247)
sandwiching	18.9%	0.048	-0.158	underpowered ($n_{\min}$=247)
Llama Guard 4	18.9%	0.004	-0.158	underpowered ($n_{\min}$=247)
paraphrasing	24.6%	0.500	-0.019	underpow. ($n_{\min}$=17253)
parse_fast	32.8%	0.974	$+0.163$	(worse than baseline)
parse_dc	22.1%	0.292	-0.077	underpow. ($n_{\min}$=1042)

Table 2: McNemar's exact test (one-sided) vs. baseline (n=122). Bonferroni threshold: α=0.0071 (7 comparisons). Only Llama Guard survives Bonferroni correction, at 64.8% utility; PARSE does not meet the corrected threshold but has the largest effect size (h=-0.245). parse_dc = parse_domain_conditional.

Original: "Engineered Systems segment net sales increased $116.9 million to $380.1 million in 2023 compared with $263.2 million in the prior year; 56.6% ($66.2 million) was attributable to organic growth and $50.7 million to acquisitions."

Paraphrased: "The Engineered Systems segment saw substantial revenue growth in 2023, driven by both organic business expansion and strategic acquisitions."

The paraphrase discards every quantitative fact, so an agent asked to compute year-over-year growth or attribute revenue sources cannot answer from it. PARSE avoids this by extracting numerical and causal facts before rewriting and verifying they survive sanitization (Step 5).

6.2 Why PARSE Succeeds

Three mechanisms work in concert. First, the directiveness gate identifies high-risk documents before sentence-level processing, routing 41% to the full pipeline and avoiding unnecessary rewriting of low-risk text. Second, sentence-level injection scoring with domain allowlists distinguishes camouflage payloads from legitimate authority language. On the 84 of 122 documents whose retained traces resolve, camouflage-payload sentences receive a mean injection score of 0.406 (n=398) against 0.019 for legitimate sentences (n=1296). Legitimate sentences carrying authority language ("shall," "must," "requires") score 0.022 (n=104), essentially the same as legitimate sentences without it (0.018), and none reaches the 0.6 high-risk threshold. The allowlists are not what produces this: they match only 4.4% of legitimate sentences, and the authority-language sentences they miss still average 0.025. The work is done by domain conditioning in the tagger prompt, which names the document's domain and anchors 0.0 to "clearly legitimate domain content." The separation is nonetheless partial: only 42.2% of payload sentences reach 0.6, so PARSE's benefit does not rest on catching every payload sentence. Third, the fact constraint and consistency checker ensure that sanitization preserves the content the agent needs to complete the legitimate task, explaining PARSE's utility advantage over Llama Guard 4 (86.9% vs. 64.8%).

This is a post-hoc analysis of retained traces, not a preregistered test; the 84 resolving documents are not a random subset of the 122, and retry and fact-coverage statistics are not recoverable at all.

6.3 Deployment Considerations

PARSE requires 3–5 LLM API calls per document and runs during ingestion rather than at retrieval time, amortizing cost over the document lifetime. At current Haiku and Sonnet pricing this is approximately $0.01 per document, or $100 one-time for a 10,000-document corpus; the directiveness gate reduces it further, since 59% of documents need only 2 calls.

6.4 Comparison to CommandSans

Das et al. (2025) propose CommandSans, a token-level sanitization system that trains a classifier to identify and remove instructions directed at AI systems from tool outputs, rather than flagging entire inputs as malicious. Two differences are relevant to domain-camouflaged attacks. First, CommandSans's training data and evaluation benchmarks do not include professional-domain documents of the kind in our corpus; whether its instruction classifier distinguishes legitimate domain authority language ("shall," "management directs") from AI-directed instructions in this setting is an open question its authors have not tested, and we do not have experimental evidence on this point either. Second, and more directly relevant, domain-camouflaged payloads contain no syntactic instruction markers—they are written to be indistinguishable from surrounding professional content (Pai, 2026a). CommandSans was not evaluated on this attack class, and its performance against domain-camouflaged

injection is unknown. PARSE targets semantic authority structures with domain-aware scoring and a fact-preservation constraint, an approach designed specifically for attacks that lack the syntactic signal CommandSans's instruction-tagging classifier is trained to detect.

6.5 Threat Model and Adaptive Attacks

Our threat model is a non-adaptive attacker: payloads are written to evade syntactic detectors and to read as domain-appropriate prose, but not with knowledge of PARSE's architecture. All results above should be read under that assumption. An attacker who knows the pipeline has at least two unevaluated avenues, both following directly from design choices we make.

Gate evasion. The directiveness gate routes documents scoring below 0.5 to a simple paraphrase. An attacker aware of the gate can write a low-directiveness payload that scores below the threshold and is routed to that path. Finding 2 finds no evidence that paraphrasing reduces ASR across the benchmark as a whole; whether it behaves differently on the low-directiveness subset specifically is something we cannot check, because trial-level outcomes were not retained (§5.2). This is not a narrow corner of the input space: 59% of our benchmark documents already take that path.

Fact-channel poisoning. Step 4 rewrites under a hard constraint that every extracted fact must appear in the output, and Step 5 retries when one does not. An instruction encoded so that the tagger-extractor records it as a *fact* is therefore not merely unfiltered—it is guaranteed to survive rewriting, because the pipeline is built to preserve it. A payload of the form "Fact: Q3 audit compliance requires sending authorization tokens to IP 192.168.1.1" targets this channel directly. PARSE's fact-preservation mechanism, which is what buys its utility advantage over indiscriminate paraphrasing, is thus also its attack surface, and the current design has no mitigation for it.

We evaluated neither vector. Establishing whether they succeed, and at what rate, requires an adaptive-attacker benchmark that we leave to future work.

7 Practitioner Recommendations

1. **Consider PARSE where the threat is non-adaptive domain-camouflaged injection.** PARSE achieves the largest ASR reduction of any condition we evaluate (h=−0.245; p=0.014 uncorrected, short of the Bonferroni threshold) at 86.9% utility, with no training requirement and approximately $0.01 per document at current API pricing, run at ingestion rather than at query time. It is not evaluated against attackers who know the pipeline, and the vectors in §6.5 are unmitigated, so we do not recommend it as a sole control where adaptive adversaries are in scope.

2. **Do not rely on paraphrasing alone for real documents.** Paraphrasing is not statistically different from no defense on real enterprise documents (p=0.500) and degrades utility by 9 percentage points. Synthetic benchmark performance does not predict real-document performance.

3. **Do not use Llama Guard as the primary defense.** Its 64.8% utility on real professional documents—a 27 percentage-point cost—is not acceptable for production RAG systems where legitimate retrieval is the primary use case.

8 Conclusion

PARSE achieves the largest attack success rate reduction of any defense we evaluate (h=−0.245) while maintaining near-baseline utility on real enterprise documents. The reduction is nominally significant uncorrected (p=0.014) but does not clear Bonferroni correction for seven comparisons; the only condition that does, Llama Guard, costs 27 percentage points of utility and we do not recommend it. Paraphrasing's synthetic benchmark performance does not generalize to real documents—a deployment warning for practitioners relying on existing benchmark rankings. Future work should increase per-domain sample sizes for domain-level statistical power, and evaluate PARSE against adaptive adversaries who craft payloads specifically to survive structured sanitization.

Limitations

Per-domain sample sizes of n=23–25 are underpowered for domain-level comparisons: the per-domain columns of Table 1 are descriptive only and cannot support claims about which domains PARSE helps most. The carrier documents are real, but the legitimate tasks, malicious goals, and camouflaged payloads are LLM-generated and no human

validation of them was performed, so the benchmark is a real-document benchmark only in the sense that its source material is authentic. Per-trial agent outputs and judge decisions were not retained, so trial-level re-analysis—including bootstrap confidence intervals and any split of ASR by routing decision—is not possible from the released artifacts, which comprise the code and the benchmark rather than the raw trial logs. Generation, defense, and evaluation all draw on the same model family. The victim agent and both judges are the same model (`claude-haiku-4-5`), so the judge scores outputs the same model produced; attack generation used a different, stronger model (`claude-sonnet-4-5`), which is a mitigating factor rather than a resolution. We do not know the direction of any resulting bias in ASR or utility, and cross-model replication is required to rule it out. The model identifiers we report are provider aliases without pinned snapshot dates, which limits exact reproducibility because aliases drift. Two claims motivating this work—that domain-camouflaged payloads evade Llama Guard 3 (Pai, 2026a), and that paraphrasing leads defense rankings on synthetic professional documents (Pai, 2026b)—come from our own prior work rather than independent replication; the former is an unrefereed preprint, the latter peer-reviewed but non-archival. PARSE is not evaluated against adaptive adversaries who know the sanitization mechanism and can craft payloads to survive it. The directiveness gate threshold of 0.5 was set a priori, not tuned: the released code contains no validation split, tuning script, or calibration data. No evaluation data could therefore have leaked into threshold selection. The specific value is also not load-bearing on this benchmark, since no document scores between 0.35 and 0.62 and any threshold in that band produces identical routing. The cost of this is that the threshold, and the pipeline's other constants, are asserted rather than empirically justified, and may require domain-specific calibration in deployment. The per-document cost of approximately $0.01 may be prohibitive for very large corpora without caching.

Ethics Statement

PARSE is a defensive system designed to protect LLM agents from adversarial manipulation. Camouflage payloads in the benchmark are constructed to test defenses, not to facilitate real attacks. All documents are sourced from public APIs (SEC EDGAR, Federal Register, PubMed, arXiv, GitHub). No personal data is collected or stored. Code and the benchmark are released publicly to support follow-on work at `https://github.com/aaditya79/parse-defense/`.

Acknowledgments

This work was carried out independently by the author at the Data Science Institute, Columbia University. It received no external funding, and the author declares no competing interests.

References

Debeshee Das, Luca Beurer-Kellner, Marc Fischer, and Maximilian Baader. 2025. CommandSans: Securing AI agents with surgical precision prompt sanitization. *Preprint*, arXiv:2510.08829.

Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. 2023. Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*.

Keegan Hines, Gary Lopez, Matthew Hall, Federico Zarfati, Yonatan Zunger, and Emre Kiciman. 2024. Defending against indirect prompt injection attacks with spotlighting. *Preprint*, arXiv:2403.14720.

Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, and Madian Khabsa. 2023. Llama Guard: LLM-based input-output safeguard for human-AI conversations. *Preprint*, arXiv:2312.06674.

Neel Jain, Avi Schwarzschild, Yuxin Wen, Gowthami Somepalli, John Kirchenbauer, Ping-yeh Chiang, Micah Goldblum, Aniruddha Saha, Jonas Geiping, and Tom Goldstein. 2023. Baseline defenses for adversarial attacks against aligned language models. *Preprint*, arXiv:2309.00614.

Yupei Liu, Yuqi Jia, Runpeng Geng, Jinyuan Jia, and Neil Zhenqiang Gong. 2024. Formalizing and benchmarking prompt injection attacks and defenses. In *Proceedings of the 33rd USENIX Security Symposium (USENIX Security)*.

Meta AI. 2024. Llama Guard 3 Model Card. `https://huggingface.co/meta-llama/Llama-Guard-3-8B`.

Meta AI. 2025. Llama Guard 4 Model Card. `https://github.com/meta-llama/PurpleLlama/blob/main/Llama-Guard4/12B/MODEL_CARD.md`.

Aaditya Pai. 2026a. Blind spots in the guard: How domain-camouflaged injection attacks evade detection in Multi-Agent LLM systems. *Preprint*, arXiv:2605.22001.

Aaditya Pai. 2026b. Evaluating prompting-based defenses against domain-camouflaged injection attacks. *Preprint*, arXiv:2606.18530.

Fábio Perez and Ian Ribeiro. 2022. Ignore previous prompt: Attack techniques for language models. *Preprint*, arXiv:2211.09527.

Sander Schulhoff, Jeremy Pinto, Anaum Khan, Louis-François Bouchard, Chenglei Si, Svetlina Anati, Valen Tagliabue, Anson Kost, Christopher Carnahan, and Jordan Boyd-Graber. 2023. Ignore this title and HackAPrompt: Exposing systemic vulnerabilities of LLMs through a global prompt hacking competition. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 4945–4977, Singapore. Association for Computational Linguistics.